

NYU TANDON · MECHANICAL ENGINEERING SENIOR DESIGN · PROJECT 19

A Soft Robotic Assistant for Physical Interaction with Elderly and Rehabilitation Patients

Force you can measure. Bounded when the power is off.

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THE PITCH IN ONE LINE

Rehab is rationed by therapist time — and the machines that could extend it are too rigid to leave near a frail patient.

Our answer is a soft arm a patient can lean on, which measures and reports the force it applies at every moment of contact.

Who our customer is

BUYER

Clinics and care facilities

Physical therapy clinics and elder-care facilities. Short on therapist hours, not on patients.

END USER

Patients

Elderly and rehabilitation patients who need steadying, guided reach and postural support — hundreds of times, not once.

END USER

Therapists

One therapist covers several patients. A passive gravity-balancing orthosis produced sustained gains on about 4 minutes of therapist attention per hour.

Patients need repetitions. Clinics need them delivered without a therapist inside every one.

ASTRA · Project 19 · Soft Robotic Assistant



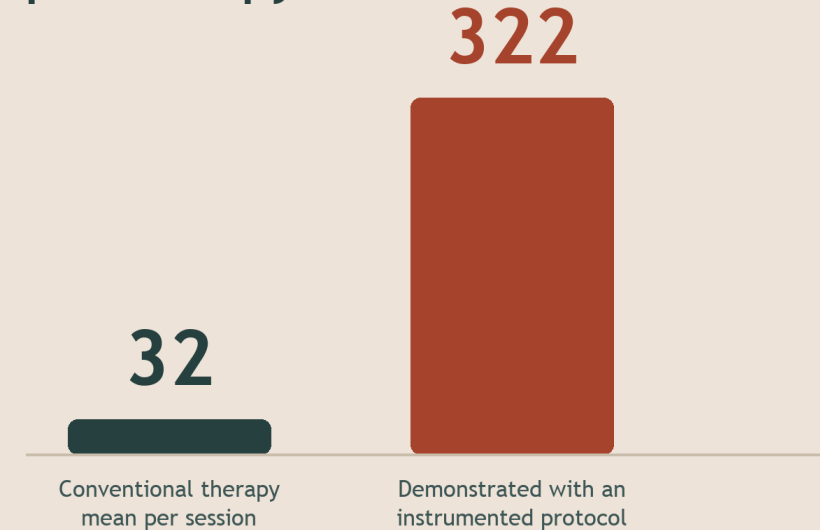
What their problem is

Recovery runs on repetitions, and every repetition costs a therapist's hands.

Conventional therapy delivers a mean of 32 upper-limb repetitions per session (95% CI 20–44). An instrumented protocol has reached 322 in the same hour.

The ceiling on therapy is staff time, not patient willingness — and the machines meant to lift that ceiling are either unsafe to leave against a frail body, or cannot carry load at all.

Upper-limb repetitions
per therapy session



**The patient needs contact that is both load-bearing and safe.
Nothing on the market is both.**

Why nobody has already fixed it

Log scale · source: docs/prior-art.md



A standards vacuum

ISO 13482 publishes no contact limits, and ISO/TS 15066 states it does not apply to non-industrial robots — and the pain data behind both comes from healthy young volunteers.

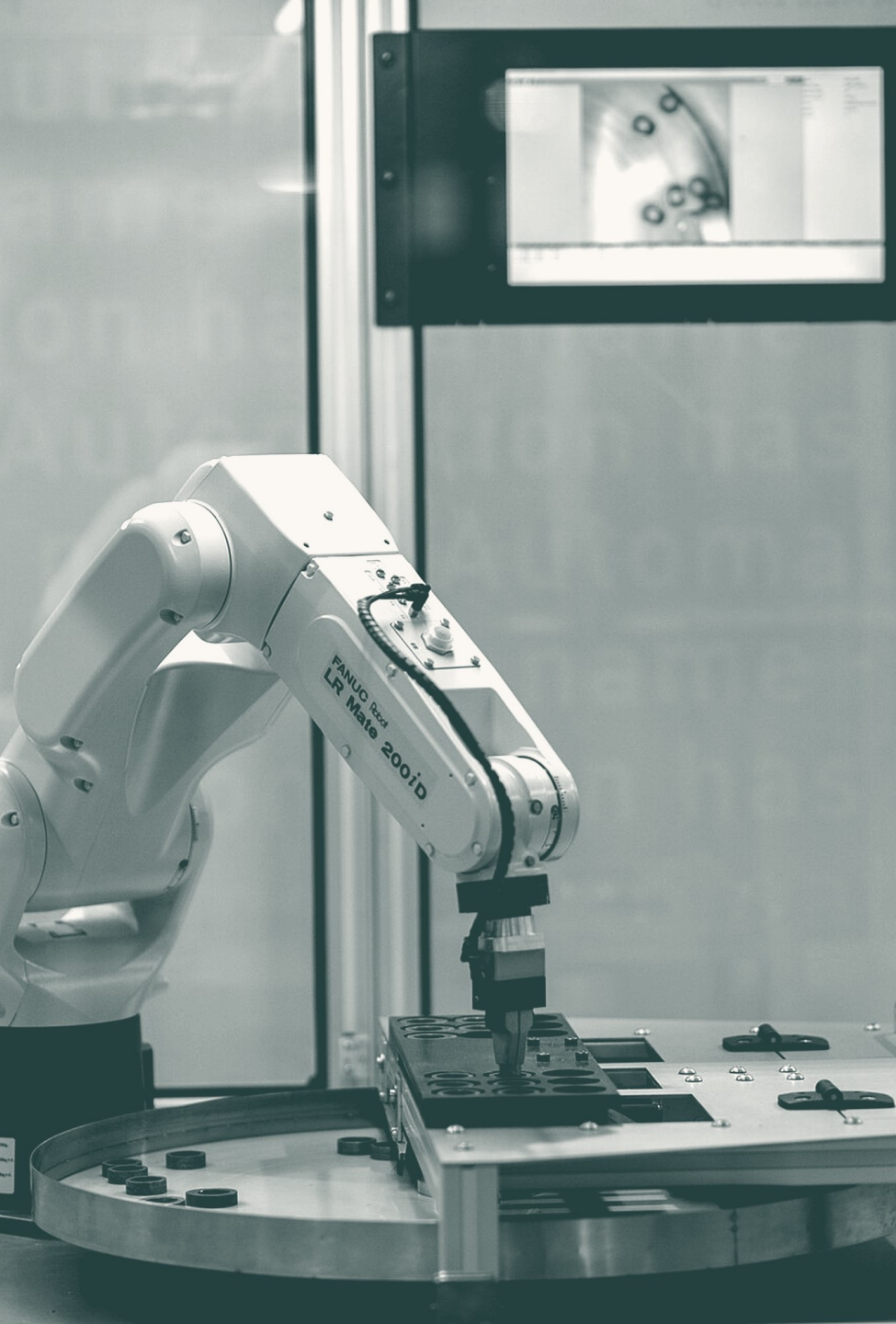
The wrong population

Pressure-pain threshold drops significantly over age 62, and reduced protective sensation travels with reduced tissue tolerance. Pain onset is not a valid proxy for injury onset here.

What exists today, and what it never reports

Compare	Rigid patient-handling (RIBA, RoNA)	Soft freestanding humanoids (HuggieBot, King Louie)	Worn exosuits (Harvard soft shoulder)	ASTRA
Assistive force	Lifts 60–80 kg	None published	6.6 N·m per limb, measured on patients	28 N, measured and reported
Load path	Rigid to floor, on a 140–230 kg robot	To floor, unquantified	Onto the patient's own skeleton	Rigid to floor, compliance in the actuation
Status	Discontinued — RIKEN-SRK closed 2015, none deployed in care	48 human subjects in uncontrolled full-body contact; force never reported	Delivering measured assistance to real patients	Bench instrument, supervised use

The soft freestanding torso is not empty ground. What nobody has built is the instrument.



VALUE PROPOSITION

A compliant support arm that delivers a characterised, instrumented contact-force envelope – measured, not asserted.

— Unlike rigid assistive robots

A series spring makes interaction force readable from its own deflection, and bounds that force when the actuator is unpowered.

— Unlike the soft humanoids already built

The deliverable is the measurement, not the demonstration. We publish the envelope, the method, and the conditions it holds under.

What we will actually build

ARCHITECTURE

Rigid spine, soft interface

Rigid grounded load path, compliance placed in the actuation, designed soft covering at the contact patch.

Series-elastic over variable-stiffness; impedance control, not admittance.

RULED OUT ON THE NUMBER

Not a fully inflatable arm

$M = P\pi D^3/8$ gives about 26 N·m for a Ø150 mm arm at 20 kPa.

A 200 N lean at 0.5 m applies roughly 100 N·m. The inflatable branch fails by a factor of four.

BEHAVIOUR

Stiffens, holds, then releases

Compliant on approach, stiffer under load, with an explicit braked state for a sustained lean.

Removing power reduces the force it can apply rather than trapping the user.

Material down-select — silicone, TPU, pneumatic bladder, tendon drive — runs against 28 N and ≈ 10 N·m, not the tens of N·m the brief implies.

Scope: what we will have by the end of the year

1 • DESIGN

Down-select the actuator

Against 28 N of assistive force and ≈ 10 N·m of shoulder gravity compensation.

At that target, actuator families that look disqualified come back into range.

2 • BUILD

One instrumented arm

A series-elastic support arm on a base that reacts ≥ 330 N applied at hand height without tipping.

3 • TEST

Measure the envelope

Force output, deflection-based force estimate, contact pressure, unpowered residual force — plus a human-subject approachability study.

Out of scope, deliberately: sit-to-stand lift (>121 N·m at the hip, which a commercial lower-limb exoskeleton already fails), unsupervised operation, and contact testing with frail participants. This is a bench instrument, not a clinical product.

How we will know it worked

Measure	How	Target
Assistive force	Load cell at the contact point, compared against force estimated from spring deflection	28 N sustained, ± 6 N modulation band
Contact safety	Quasi-static clamping test and pendulum impact rig, sampled ≥ 10 kHz with a 100 Hz filter	Below 50 N/cm ² at the sternum (Behrens, n = 112 — not ISO's 120), margin no tighter than 30%
Unpowered bound	Cut power under a held lean; record residual force and whether the arm releases	Bounded, non-clamping, releases
Approachability	Within-subjects, soft covering on vs off, same arm and motion. Perceived Danger as primary measure, RoSAS for approachability, plus stop distance in cm	n = 32–40 healthy adults, preregistered

Healthy adult volunteers at the Robotics Center, supervised. Contact testing with frail participants is full-board IRB and outside this calendar — we start the paperwork with our senior design professor either way.

MAIN MESSAGE

Compliance is not what makes it safe. Measurement is.

We will produce the first instrumented contact-force envelope for a soft assistive arm: 28 N, read from spring deflection, bounded when the power is off.

Thank you. Questions?

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